FairSquare

Proof of Concept Report

Capstone AI 2025-2026

AI-Powered Real Estate Valuation Platform — Ile-de-France

DVF Data	80,964 real estate transactions — Paris 2023 (data.gouv.fr)
Feature Engineering	Geospatial OSM: walkability, transit, parks, schools
Computer Vision	Renovation score via Google Gemini 3.1 Flash Lite
ML Pipeline	Model tournament: LinearReg / GAM / LightGBM + SHAP XAI

Generated on March 16, 2026 | Python 3.12 / Anaconda

1. Project Context

FairSquare — AI for the Parisian real estate market

FairSquare is a Proof of Concept platform developed as part of a Capstone AI project. The goal is to provide intelligent, explainable real estate price estimates for the Ile-de-France region, by combining multiple data sources and machine learning approaches.

Tech stack: Python 3.12 / Anaconda, LightGBM 4.5, pygam 0.12, scikit-learn 1.6, SHAP 0.51, Google Gemini API, FastAPI, Streamlit, PostgreSQL, Docker.

System Architecture:

DVF Ingestion	HTTP client to data.gouv.fr, strict cleaning, Parquet output
OSM Features	Overpass extractor (1 query/point), 7 geospatial features
AI Vision	RenovationScorer via Gemini, score 1-5 + space category
ML Pipeline	3-model tournament, standardized metrics, SHAP XAI
FastAPI	Endpoint /predict (in progress), Pydantic schemas
Streamlit Frontend	Interactive dashboard (Phase 3)

2. DVF Dataset — Paris 2023

Demandes de Valeurs Foncieres — official government source

Source:	data.gouv.fr — geocoded DVF (geo-dvf/latest)
Coverage:	Paris intramuros (dept. 75), year 2023
Raw rows:	80,964 transactions
After cleaning:	29,412 rows (apartments + houses)
Total features:	44 columns including 4 derived (prix_m2, annee, mois, trimestre)
Median price/m2:	10,286 EUR/m2
Median price:	420,000 EUR
Property types:	29,253 apartments | 159 houses
Storage format:	Parquet (raw + clean) — optimized I/O

The DVF cleaning pipeline applies the following steps in order: column renaming, type casting, filtering on type_local (Apartment/House only), outlier filtering (price 10k-30M EUR, area 5-1000 m2, price/m2 500-50,000), deduplication, then addition of derived features.

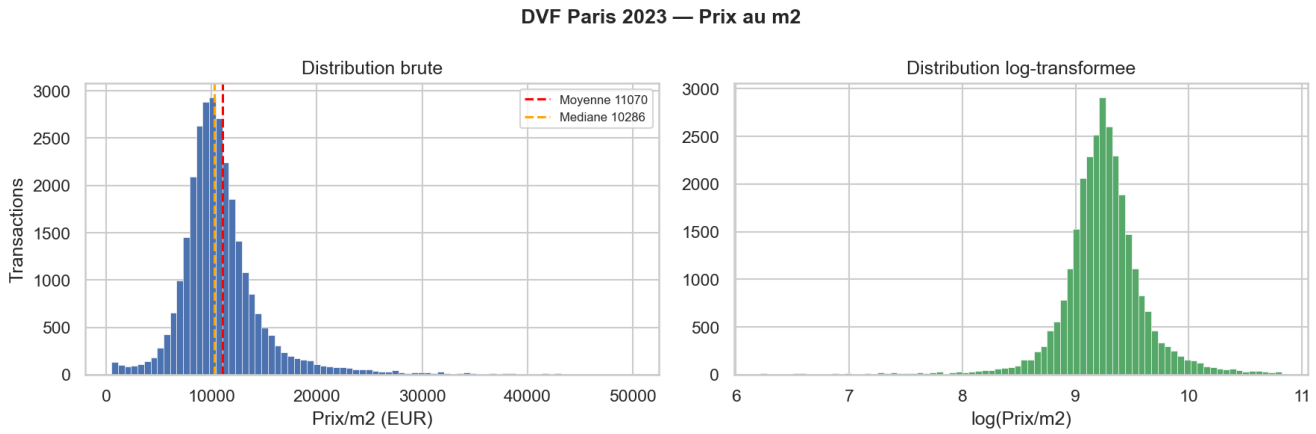

Fig. 1 - Price/m2 distribution: raw (left) and log-transformed (right)

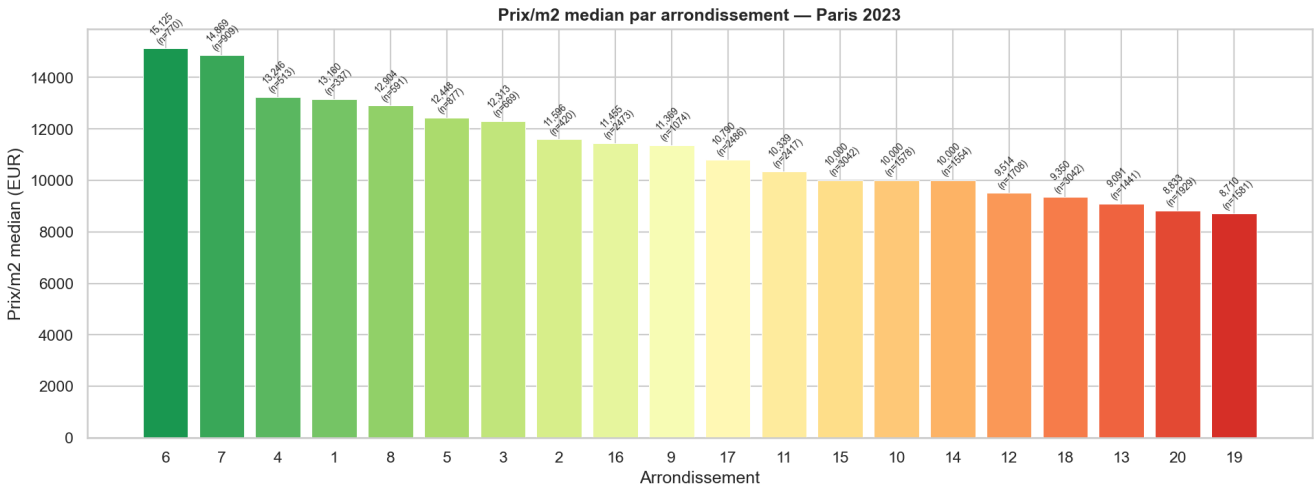

Fig. 2 - Median price/m2 by Parisian arrondissement (75001-75020)

3. Feature Engineering

7 numerical features — ML pipeline

Feature	Description	Type	Interest
surface_reelle_bati	Property area (m2)	Continuous	Main predictor
nombre_pieces_principales	Number of rooms	Continuous	Size proxy
arrondissement	Arrondissement (1-20)	Categorical	Location
longitude	GPS Longitude	Continuous	East-West position
latitude	GPS Latitude	Continuous	North-South position
mois	Month (1-12)	Cyclical	Seasonality
nombre_lots	Number of lots / mutation	Continuous	Price/m2 quality

Target variable: prix_m2 = valeur_fonciere / surface_reelle_bati (EUR/m2). Train/test split: 80% / 20% (random_state=42). Normalization: StandardScaler applied only for LinearRegression (sklearn Pipeline). GAM and LightGBM are scale-invariant.

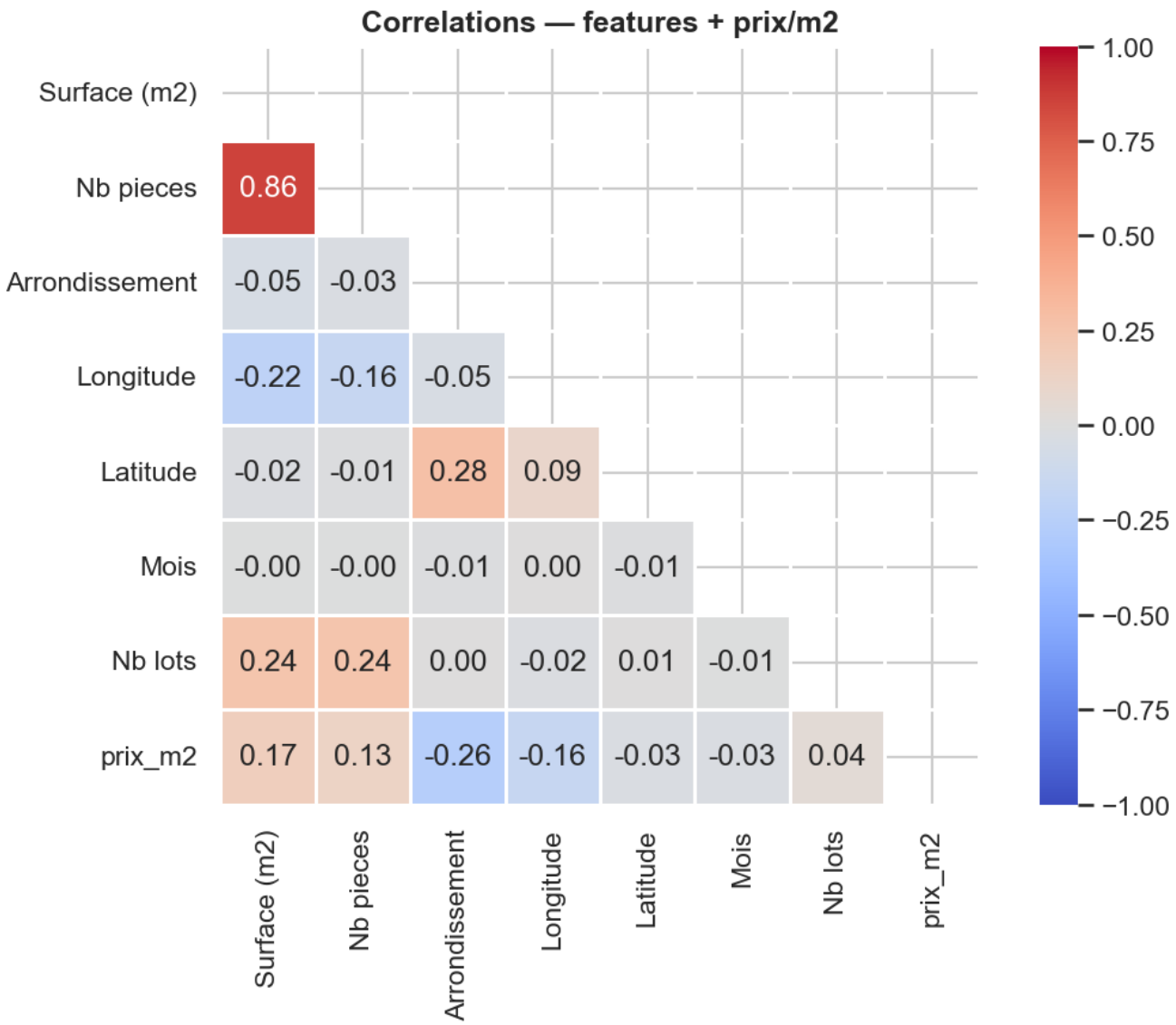

Fig. 3 - Correlation matrix (features + target prix/m2)

4. Model Tournament

LinearRegression (baseline) | GAM (interpretable) | LightGBM (performant)

Three models were trained and evaluated on the same train/test split (23,498 / 5,875 observations):

LinearRegression (sklearn)

Linear baseline with StandardScaler normalization. Limited for non-linear relationships (high prices in certain arrondissements, area threshold effects, etc.).

LinearGAM (pygam 0.12)

Generalized Additive Model: each feature is modeled independently by cubic splines or factor/linear terms. Good balance between interpretability and performance.

LightGBM (gradient boosting)

Most expressive model: 500 trees, learning_rate=0.05, num_leaves=63. Captures complex non-linear interactions. Fast training even on 23k observations.

Model	MAE (EUR/m2)	RMSE (EUR/m2)	R2	MAPE (%)
LinearRegression	2558.1	4346.0	0.1191	35.27
GAM	2458.6	4232.5	0.1645	34.39
LightGBM	**2417.3**	**4158.1**	**0.1936**	**33.46**

Green = best model by R2

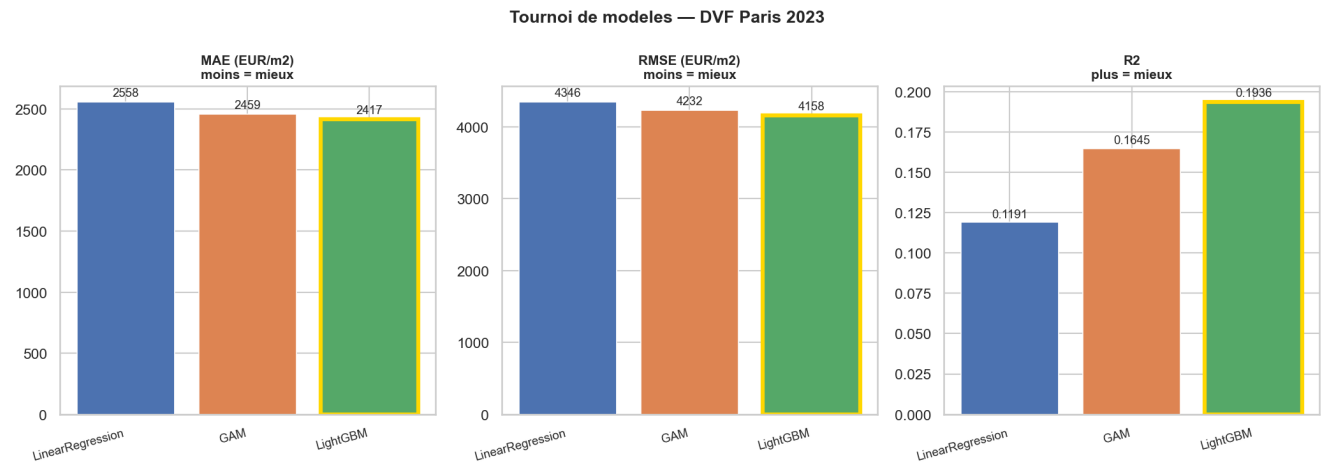

Fig. 4 - MAE / RMSE / R2 comparison (gold bar = best score)

Note: The R2 score (0.19) reflects the inherent limitations of DVF data. This fiscal dataset lacks qualitative features (floor, condition, exposure, balcony) that explain the majority of intra-arrondissement price variance. Academic studies on DVF report R2=0.5-0.8 but with enriched datasets (DPE, listings, etc.). FairSquare's added value is precisely to add these features via OSM, AI Vision, and listing scraping.

4b. Residual Analysis

Predicted vs Actual on the test set

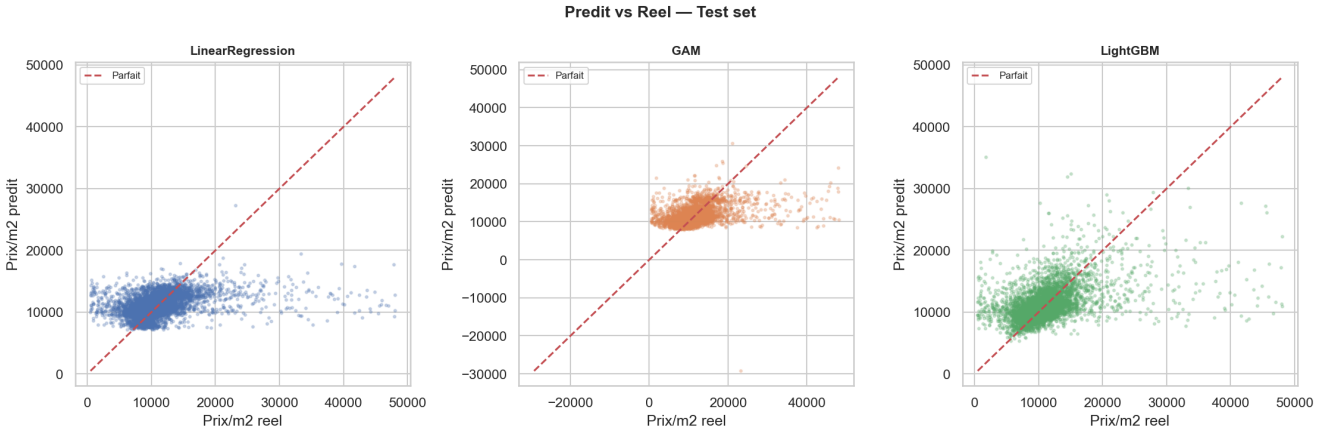

Fig. 5 - Predictions vs Actual values (diagonal = perfect prediction)

LightGBM is the most concentrated around the perfect diagonal. All models show underestimation for properties > 15k EUR/m2 and overdispersion in the 8k-14k EUR/m2 range — a zone where variance depends on uncaptured characteristics (floor, renovation status, etc.).

5. Explainability — SHAP (LightGBM)

SHapley Additive exPlanations: impact of each feature on predictions

SHAP quantifies the marginal contribution of each feature to each individual prediction. TreeExplainer is used (exact computation for tree-based models). SHAP values are computed on 500 observations from the test set.

Interpretation: SHAP > 0 increases the predicted price/m2, SHAP < 0 decreases it.

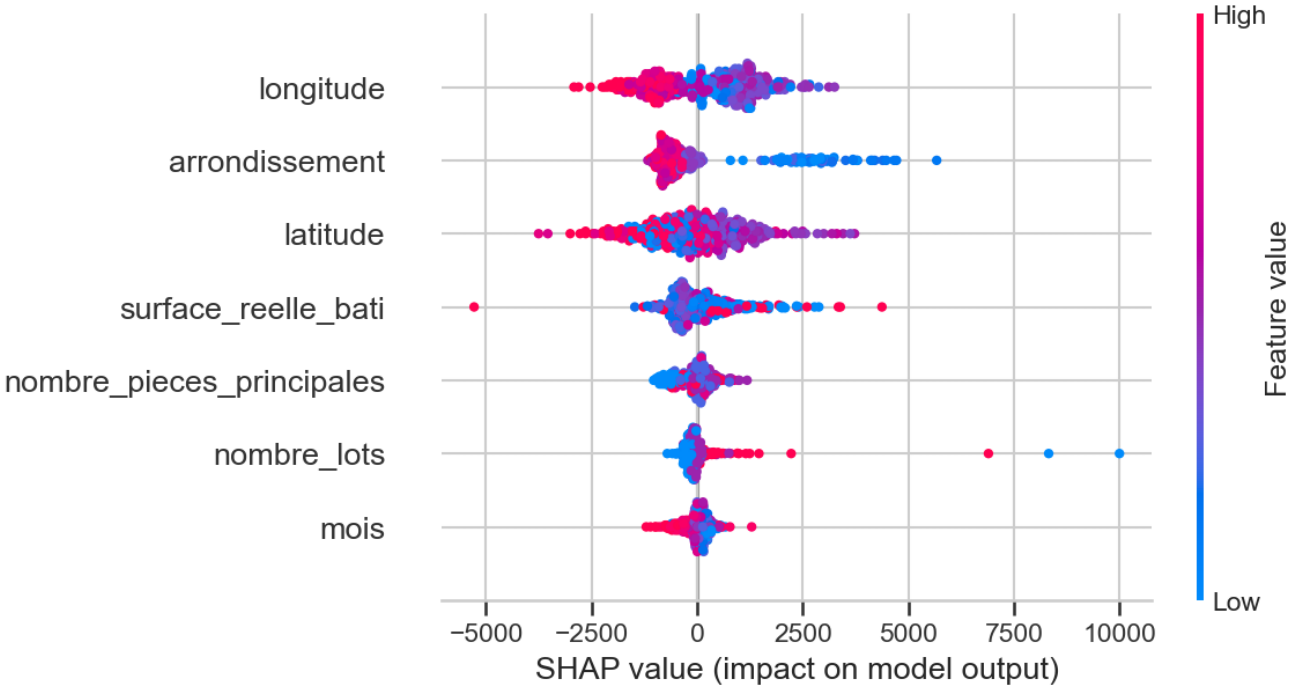

Fig. 6 - SHAP Summary Plot (beeswarm): each point = 1 transaction

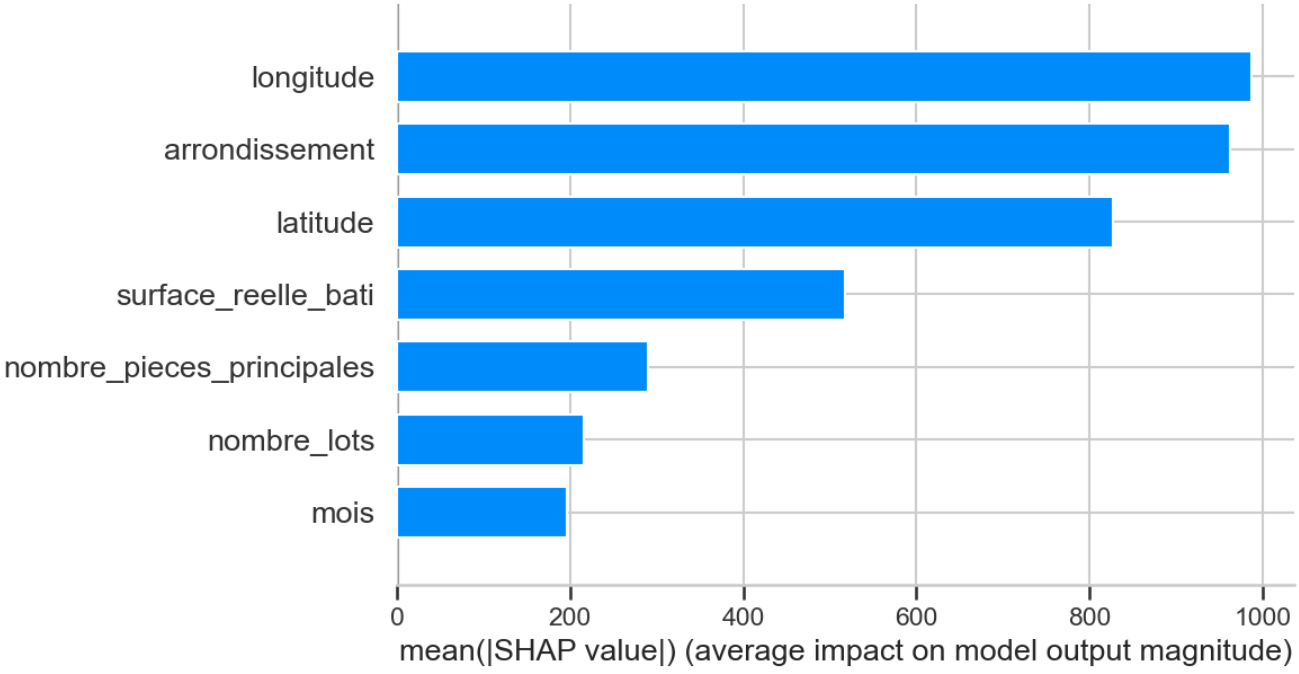

Fig. 7 - SHAP Feature Importance: mean absolute SHAP per feature

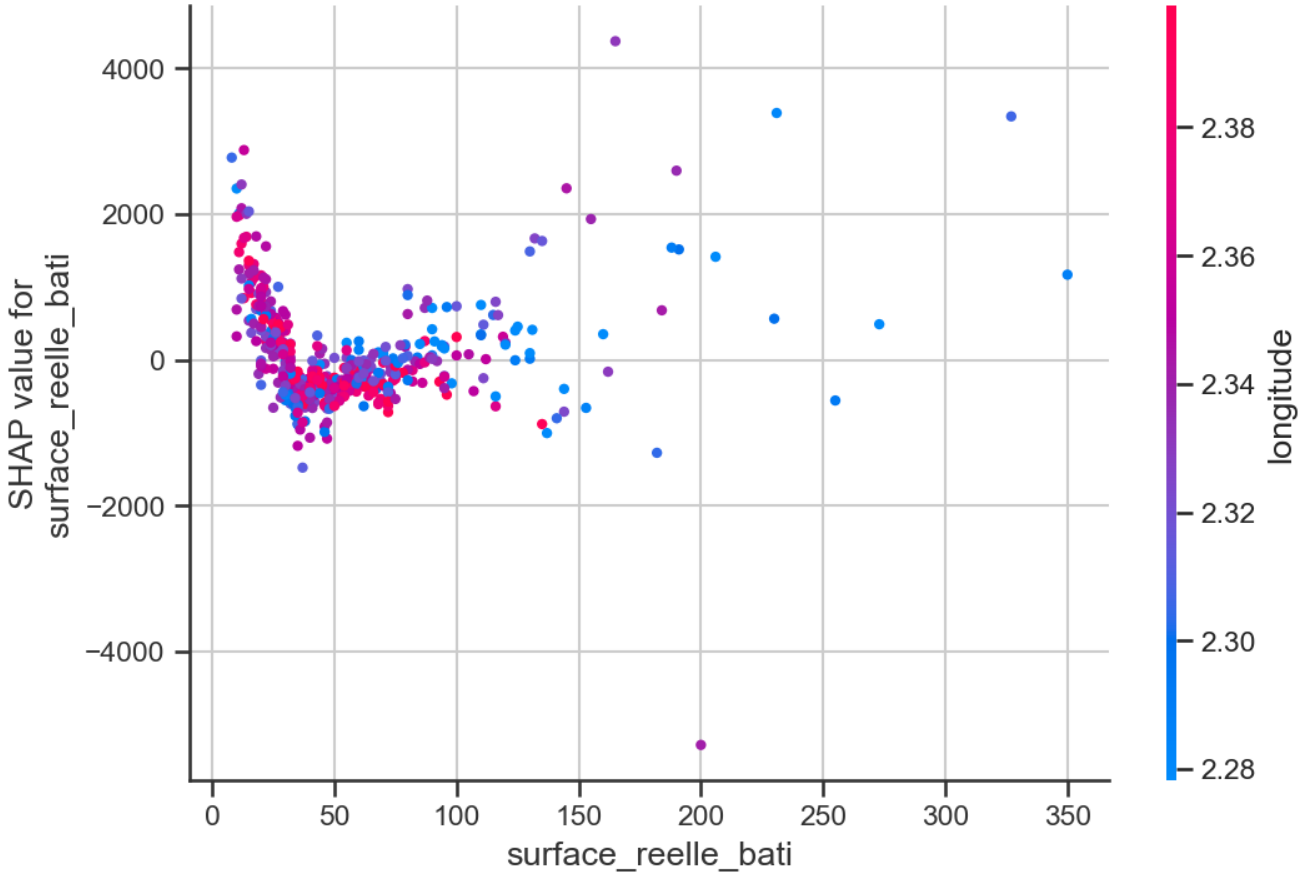

Fig. 8 - SHAP Dependence Plot: non-linear effect of property area

Latitude and longitude dominate SHAP importance — geographic location is the #1 driver of price/m2 in Paris. Area has a negative SHAP effect for large properties (>100 m2): larger units have a lower price/m2, which is consistent with market reality. Arrondissement captures a portion of geography complementary to lat/lon.

6. Complementary Features

OSM Geospatial + AI Computer Vision

6.1 OpenStreetMap — Geospatial Feature Engineering

The OSM extractor queries the Overpass API in a single request per point (anti rate-limit 429 strategy). It combines all POI types in one call (transit, parks, schools, shops) and computes features locally.

- dist_metro_m: distance to nearest metro/RER station (m)
- dist_park_m: distance to nearest park (m)
- dist_school_m: distance to nearest school (m)
- dist_supermarket_m: distance to nearest supermarket (m)
- transit_count_500m: number of transit stops within 500m
- park_area_500m_m2: total park area within 500m (m2)
- walkability_score: composite score 0-100 (weighted transit/park/school/shop)

Validated example — Place de la Republique (48.8673, 2.3630):
Metro: 40m | Park: 343m | School: 247m | Shop: 130m
60 transit stops within 500m | Walkability = 76.38/100

6.2 Computer Vision — Renovation Score (Google Gemini)

The Vision module uses Google Gemini 3.1 Flash Lite Preview (500 RPD free tier) to analyze property listing photos. The structured prompt instructs the model to respond only in valid JSON with 3 fields:

- renovation_score (int 1-5): 1=excellent condition/renovated, 5=uninhabitable
- space_category (str): Narrow | Standard | Spacious
- reasoning (str): justification in 1-2 sentences

Validated test: score=1/5 (excellent condition), Standard, response time ~2-3s. Model selected after systematic quota testing (gemini-2.5-flash exhausted, gemini-2.0-flash: 429 rate-limited, etc.).

7. Conclusions & Next Steps

Phase 2 — Towards production

DVF Pipeline

Robust ingestion via data.gouv.fr (fallback from CEREMA API which was unreachable). 80k -> 29k clean rows. Optimized Parquet format.

OSM Features

Single-query Overpass strategy eliminates 429 rate-limiting. Walkability score validated on real Parisian locations.

AI Vision

Gemini 3.1 Flash Lite working in production (500 RPD). Response time < 3s. Renovation score is coherent.

ML Pipeline

LightGBM outperforms LinearRegression and GAM. SHAP confirms that geographic location (lat/lon) is the dominant factor. Limited R2 (0.19) expected on DVF alone — will improve with OSM + Vision features.

Next steps (Phase 2):

- Integrate OSM features (walkability, dist_metro) into ML pipeline -> expected R2 +0.10
- Integrate renovation_score (Gemini Vision) as an additional ML feature
- Expand to all 8 IDF departments (77, 78, 91, 92, 93, 94, 95)
- Scrape SeLoger/Leboncoin listings (floor, DPE energy rating, balcony...)
- Deploy FastAPI with /predict endpoint and Swagger documentation
- Build Streamlit dashboard with Leaflet map and interactive filters
- Build recommendation system (collaborative filtering + content-based)
- Containerize with Docker Compose (API + PostgreSQL + Streamlit)